

Divyanshu Singh

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EDUCATION

Texas A&M University, *College of Engineering, College Station*

• **Master of Science, Computer Science** | **GPA: 3.85**

May 2026

• **Research Focus:** Machine Learning aided Computer Vision for Human 3D Reconstruction in Dynamic Scenes with Multiple Bodies

• **Bachelor of Science, Computer Science** | **Minor:** Mathematics

December 2024

SKILLS

Programming Languages: JavaScript, TypeScript, Python, GoLang, C/C++, MATLAB, SQL, Haskell

Frameworks: Flutter, Spring/Spring Boot, React.js, Express.js, Flask, Go-Chi

Tools: GitHub, Linux/Unix, SSH, AWS (Cloud Practitioner), Google Cloud Platform, Docker, Kubernetes

Databases: PostgreSQL, MongoDB, Redis

RELEVANT EXPERIENCE

Student Software Engineer, *Texas A&M University College Station, TX*

Feb. 2024 - Present

- Led full-stack development of [DataScribe.Cloud](#), a research data platform adopted by teams at **multiple universities and national labs**. Primary developer from MVP to production
- Built and deployed an LLM-powered research agent with tool-use capabilities including **RAG-based** materials science **Q&A** and **dynamic ML workflow orchestration**, enabling researchers to query and trigger complex pipelines in natural language
- Architected automated CI/CD pipeline (Docker + Kubernetes) eliminating manual image builds and registry pushes. Reduced deployment from 2 hours of manual container/manifest updates to a fully automated ~5-minute process

Software Engineering Intern, *Exabeam Inc. Foster City, California*

May - Aug. 2023

- Built a **Spring Boot REST API** cost-analysis tool that gave developers real-time visibility into the compute cost impact of code changes, used by the data engineering team
- Designed and deployed **Prometheus + Grafana** observability from scratch for the core threat scoring engine, establishing the **first real time visibility** into pipeline health and enabling incident detection that previously wasn't possible
- Conducted security audits across **15+ repositories**, identified and remediated critical/high-severity **CVEs** reducing attack surface

Teacher Assistant, *Texas A&M University College Station, Texas*

Spring 2024 - Present

- C++, Quantum Computing, Programming Languages | ~150 students/semester
- Automated assignment grading pipeline with **Bash scripts**, cutting TA grading time by an estimated 60% per coding assignment cycle

PROJECTS

Vihayas (Co-Founder) [Link](#)

- Building a **new IaC** platform unifying **AWS, GCP, and Azure** under a single JSON-like configuration syntax, extensible to additional cloud infrastructure management suite and tooling
- Developed AI-powered Kubernetes management module with smart **HPA/VPA scaling**, with resource recommendations.
- Cross-platform catalog service for browsing available VM, DB, and compute configs with an agent that generates full infrastructure solutions from **natural language requirements**

PharmFindr (Co-Founder) [Link](#)

- Built microservices backend in **Go** with **PostgreSQL**, **MongoDB**, and **Redis**, aggregating FDA 483 data and cGMP compliance records into a searchable platform
- Built an internal AI agent pipeline to automate data capture and enrichment across **500+ CDMO listings**, enabling product launch within **one month** of backend completion

Ipelint MVP 2: AI-Powered Patent Analysis (Python, TensorFlow/PyTorch, NLTK, Scikit-learn, AWS Bedrock) [Link](#)

- **Fine-tuned NLP models** for patent prosecution analysis: keyword/anti-keyword identification, topic modeling, and prior art search — collaborated directly with **IP attorneys** to validate outputs
- Built a **Computer Vision model** for patent diagram analysis to surface relevant prior art, integrating with the core patent search pipeline

PUBLICATIONS

DataScribe: An AI-Native, Policy-Aligned Web Platform for Multi-Objective Materials Design *Digital Discovery*, 2026

- AI-native cloud platform unifying experimental data through knowledge graphs, FAIR-compliant metadata, and multi-objective Bayesian optimization for closed-loop materials discovery workflows.

TRAM-MP: Multi-Person Trajectory and Motion Reconstruction from Videos in the Wild *ArXiv Preprint*, 2025

- Feed-forward method for reconstructing global trajectories and body motions of multiple people from monocular videos. Achieves comparable error reduction in trajectory estimation as compared to leading models with speedup transformer-based depth estimation.

Efficient Parallelization of Point Cloud Registration: Spatial Indexing and Multi-Resolution *ArXiv Preprint*, 2025

- Demonstrates significant speedup in classical ICP algorithms through CPU parallelization, multi-resolution refinement, and spatial indexing without specialized hardware.